

# GAUGE: How Large Language Models Recover Mobility Descriptors from Individual Records

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## Abstract

Mobility science compresses a person's movement into descriptors such as radius of gyration, the count of distinct places, and the longest single jump. Mobility models take those descriptors as generation constraints, as routing inputs, and as scoring functions over populations, and never ask a model to recover one from a record. GAUGE is a benchmark that reverses the direction and measures what comes back. Across five open-weight models and 34,200 questions over individual location records, recovery splits by descriptor class rather than by difficulty. Asked to state a value directly under greedy decoding, spatial-geometric descriptors never clear their baseline, at any record length or tolerance we score. Count, sequence and retrieval descriptors do. GPT-4o, scored on a matched 960-question subset, clears no geometric cell either. The profile is bimodal in record length: 89% of model-family pairs either clear their baseline on the longest record we test or clear it nowhere. A mean accuracy therefore describes no pair. What binds recovery is the amount of record rather than the state an answer requires, established by two designs that vary different quantities, and the profile survives changes to wording, scaffolding and answer format. The failure is in carrying the computation out rather than in knowing what it is: asked for a Python expression instead of a value, three of the five models write programs that compute the same descriptors correctly on 50 records they never saw. A permutation of the true values across people reproduces every distributional score we compute while destroying every individual, and the models order people on geometric quantities in 75 of 100 settings while stating no individual's value. We give the field a second axis for evaluating mobility foundation models: recovery from one person's record, with exact answers and exact baselines, beside the distributional scores now used. Code is available at <https://github.com/mustafasameen/GAUGE>.

## CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → *Information systems applications*.

## Keywords

human mobility, foundation models, evaluation, benchmark, large language models

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## 1 Introduction

Large-scale positioning data has made it routine to describe a person's movement through a small set of summary quantities. Radius of gyration measures how far someone ranges from their centre of mass [20], the entropy of a time-ordered location sequence bounds how predictable someone is [57], and the count of familiar places is stable enough to behave as a conserved property of an individual [2]. These descriptors are the working vocabulary of human mobility research, and they are what urban mobility foundation models are built to reproduce.

Existing evaluations of mobility foundation models use these descriptors in one direction only. Radius of gyration alone appears in six recent systems in three distinct roles: as an input constraint for conditional generation [22, 45], as a divergence-based scoring function over generated trajectories [14, 24, 37, 45], and as an input feature for routing between experts [72]. In none of them is it computed from a given record and checked against that person's truth. Figure 1 contrasts the two directions. This design leaves a gap: a model can match a radius-of-gyration distribution across a city while representing nothing about any individual's radius of gyration, and a distributional score cannot separate those two cases. Li et al. [38] make the same point about generated trajectories, observing that distributional metrics such as JSD, DTW and transition-matrix norms are "suitable for population-level alignment" but "overlook the semantic plausibility and constraint fidelity of individual trajectories", and calling for evaluation aimed at individual-level reasoning. Benchmarks for spatiotemporal language models do not close it either. STBench [40] and STARK [52] evaluate reasoning over points, regions and synthetic trajectories; CityBench [16] and USTBench [34] operate predominantly at city and system level; MobQA [4] asks semantic questions about trips rather than the quantities mobility science computes. Chu et al. [8] show that apparent spatial reasoning in language models is symbolic, arising from pattern matching over spatial language rather than geometric reasoning over continuous space, but their objects are named entities and the relations between one or two geometric primitives.

This paper measures that gap directly. We ask language models to recover mobility science's descriptors from a single person's location record, and we report what comes back. To ask the question at all we build GAUGE, a benchmark of 34,200 questions across 21

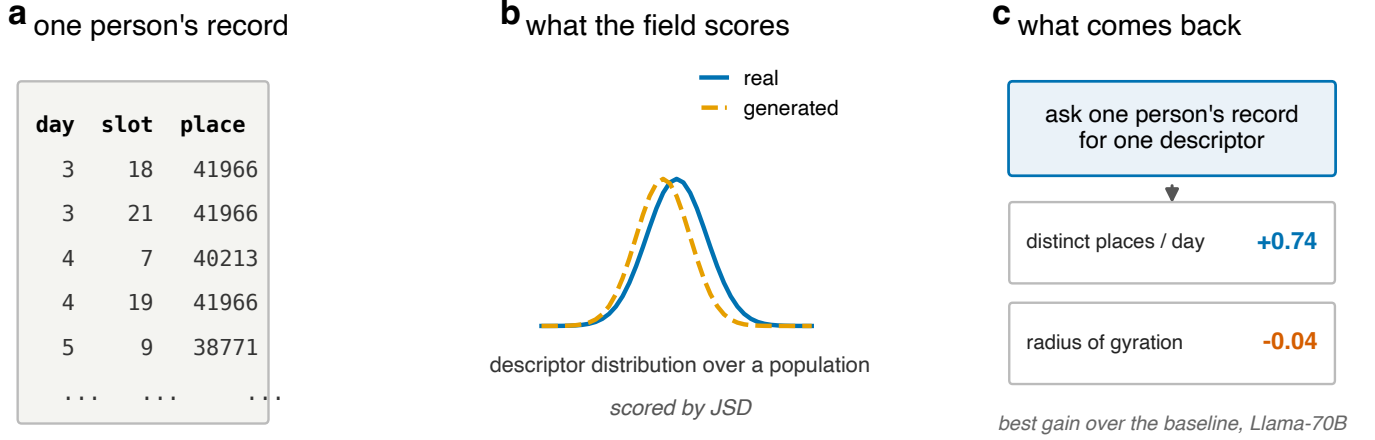


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**Figure 1: (a) The raw material: one person’s location record, a sequence of day, timeslot and place. (b) What mobility AI computes from records like it: descriptor distributions over a population, compared against real ones by divergence. (c) The direction GAUGE runs instead, and what it returns. Asking the same model for one person’s own descriptor recovers distinct places per day and does not recover radius of gyration, and which of the two it is does not depend on how hard the descriptor looks.**

descriptor families and seven record lengths. Each question targets one descriptor over one record, and a short program computes the answer from that record: both the correct answer and the baseline are exact rather than annotated. Because the questions are generated, record length is a variable we set, which lets us hold the question fixed and grow the record. We evaluate five open-weight models from 3.8B to 70B, and GPT-4o on the geometric and two contrast families. Our contribution is the measurement.

Recovery splits by descriptor class rather than by difficulty (Sections 4.1–4.3). When models state the value directly, every spatial-geometric descriptor fails to clear its baseline for every model at every record length, including under relaxed tolerances and continuous error metrics. These descriptors never had a signal to lose, and GPT-4o clears no geometric cell either. Count and sequence descriptors do clear their baselines on short records and decay as the record grows. The failure survives isolation. Handed two points and asked the distance between them, no model exceeds 2.8% exact and the best reaches 39.5% within a factor of two: the geometric step itself fails with nothing to find or aggregate. The profile is also bimodal in record length. 89% of model-family pairs either clear their baseline on the longest record we test or clear it nowhere. Detecting that a record lacks the answer collapses separately from recovering it in the three largest models. In the two smallest it never exceeds 16%.

What binds recovery is the amount of record rather than the state an answer requires (Sections 4.3–4.5). Crossing the two separates them. Recovery falls with record length at fixed required state and does not fall with required state at fixed record length, and a cross-record attribution task reaches the same conclusion from the opposite direction, improving as the same rows are divided among more records. Three changes to how we ask move the level of recovery and leave the split by descriptor class intact. Question wording does not move the geometric families, scaffolding raises recovery only where model capacity already exists, and changing the answer format to a comparison recovers graded geometric signal

on short records and loses it as records lengthen. The geometric failure also holds when the record and question carry no mobility terms, and chain-of-thought reasoning recovers only the longest jump, in the largest model, on short records (Section 4.6).

The failure is not ignorance of the descriptor. Asked for a Python expression instead of a value, Llama-3.1-70B writes programs that reproduce the reference computation on 50 held-out real records, 1,155 of 1,155, for the same descriptors it cannot state as numbers. Llama-3.1-8B writes 79 that are correct on their own record and none that survive the held-out test: correctness on the visible record alone certifies nothing (Section 4.7). Scoring recovery at three levels, distributional, ordinal and exact, separates failures that a single accuracy number reports as identical (Section 4.1).

## 2 Related Work

*Mobility foundation models.* Generative models of human movement produce next locations, full trajectories or city-scale tensors, scored against real data distributionally [14, 22, 38, 45]. They use mobility science’s descriptors as generation constraints, routing inputs and scoring functions, with radius of gyration alone appearing in six of the twenty systems of Table 1. None computes a descriptor from a given record and checks it against that person’s truth, and fifteen of the twenty expose no interface through which the question could be asked. Prediction of future behaviour from a record is a different task and does occur; what is absent is recovery of a quantity the record already determines. Li et al. [38] name this gap, and the field’s own task taxonomies carry no row for analysing one person’s record [25, 29]. The same group has since moved evaluation toward the individual. TrajGenAgent [39] scores generated trajectories with trained anomaly detectors that flag implausible visits and behavioural shifts, a check of the generated data against a learned standard of plausibility. Recovery asks the model itself for a quantity a given record already determines, scored against a computed truth.

*Spatiotemporal benchmarks for language models.* Recent benchmarks test language models on places, routes, regions and trajectories [4, 16, 34, 40, 52, 75], on sensor networks represented as graphs [49], on reasoning over long contexts [5], and on geometry supplied directly as text. Given an explicit geometry, models identify topological relations above chance [30], while apparent spatial reasoning over place names is largely pattern matching [8]. Their objects are named entities, cities, or relations between one or two geometric primitives. Ours is a quantity aggregated over one person’s longitudinal record, which no supplied geometry contains and which must be constructed before it can be reasoned about.

*Numeric representation and abstention.* Probing work finds numeracy in embeddings that degrades outside the training range [63], and locates the failure precisely: magnitude and pairwise ranking are linearly decodable from hidden states [73], numbers are held per digit so errors land in digit space [35], and a sinusoidal-basis probe decodes values almost exactly [32]. All of it uses numerals supplied in the prompt. Our quantities must be aggregated from a record first, and our evidence is behavioural. Their results predict what we see at the output: a descriptor can be ordered without being stated, and wrong answers are wrong by magnitude.

On abstention, benchmarks ask whether models know a question cannot be answered [53] and find they often do not, including where correct rejection collapses as the haystack grows [64]; retrieval systems over-refuse under noise [76] and reasoning fine-tuning degrades abstention [33]. Roig [55] reports grounding and fabrication varying independently in document question answering, with fabrication rising as context grows. In all of them unanswerability is a property of the question. Ours is a property of the record: the question is well posed and the record does not contain the answer, which the program that computes the gold can check.

*Numeric reasoning over long inputs.* Exact-state tasks degrade as sequences grow, at rates that differ by model and are not always monotone in length [12, 26, 36, 58]. Dai and Fan [10] find such collapses abrupt rather than graded, which is the shape our length profile takes. Answer position affects retrieval [44], and later work calibrates that effect [27], ties a separate bias to the spacing between relevant pieces [61], and finds its impact minimal once retrieval is realistic [9]. Our manipulation moves a single answer through absolute positions, which tests the axis Liu et al. introduced rather than the spacing axis. Producing a program instead of a value raises accuracy on numeric reasoning questions [7, 17], and on the longest-document subset of a benchmark built for that setting [74], where the target number is written somewhere in the text. Our target appears nowhere in the input, and we use the program form as a control on whether the descriptor is known at all.

### 3 Methodology

#### 3.1 Task formulation

Let a record  $R = \langle r_1, \dots, r_n \rangle$  be one person’s ordered location observations, each  $r_i$  a triple of day, timeslot and place identifier. A mobility descriptor is a function  $f(R) \rightarrow y$  that mobility science uses to summarise  $R$ , such as radius of gyration or the count of distinct places. The recovery task asks a model  $M$  for  $y$  given a

natural-language question  $q_f$  and a rendering of  $R$ , and scores  $M(q_f, R)$  against  $f(R)$ .

Descriptors differ in how much composition they require, which is the axis along which transformer failures on compositional tasks have been characterised [13], and record length is the axis along which length generalization is studied [3]. Two properties of this formulation matter. First,  $f$  is computable: the gold answer is exact rather than annotated. Second, we control  $n$  at construction time, which makes record length a manipulated variable instead of a property of whatever data was available.

#### 3.2 Question generation

We generate 34,200 questions across 21 descriptor families and record lengths from 8 to 512 rows, drawing records from YJ-Mob100K [69]. Each family is generated only at lengths where its answer distribution still varies. The grid is ragged: 108 family-by-length settings at 300 questions each, 600 for the retrieval probe. The four geometric families are generated from 32 rows upward, and only distinct places per day and recency span all seven lengths. The geometric and count families follow scikit-mobility’s individual measures [51]; the sequence, retrieval and windowed families are constructed here. Families cover five descriptor classes: spatial-geometric (radius of gyration, maximum displacement, total distance, longest jump), count and sequence (distinct places per day, transitions, two-hop paths, and the core set, the smallest number of places accounting for half a person’s records), temporal (longest stay, recency, modal timeslot), retrieval by position, and windowed counts, the distinct places among the most recent rows. Cross-record attribution is scored separately (Section 4.4).

Every question receives a per-question rigid transform and place relabelling from a pool of 512 identifiers, and we compute the gold answer after both transforms.

#### 3.3 Three-level measurement

We score three levels, because a pooled score hides distinct failure modes [46]. *Distributional* match compares the model’s answer distribution against a constant reference by Jensen-Shannon divergence [42], and counts as present when the model improves on that reference by more than .10. *Ordinal* agreement is Spearman correlation [59] against the true ordering of people, and counts as present at  $\rho \geq .20$ . *Exact* recovery is normalized gain over a per-setting baseline, present at a gain of at least .05,

$$g = \frac{\text{acc} - \text{baseline}}{1 - \text{baseline}},$$

where the baseline is the accuracy of always answering that setting’s modal value. The .05 presence bar is the one every null below is tested against; Table 2 bolds a weaker one, a 95% interval excluding the baseline, which admits small positive effects the presence bar rejects. Because  $f$  produces every gold answer, the baseline is exact rather than estimated, and a model that answers the modal value scores  $g = 0$ . Stating a baseline is one of the 46 benchmark best practices Reuel et al. assess [54], and the baselines vary widely across our settings. Reporting accuracy alone would confound a descriptor’s difficulty with the concentration of its answer distribution.

**Table 1: Capabilities of twenty mobility and urban foundation models. The first four columns are widely shared across these systems. No system emits a descriptor value for an individual and none evaluates recovery of one, the gap this paper measures.**

| System                       | Individual record | Language interface | Raw location input | Multiple tasks | Descriptor-valued output | Recovery evaluated |
|------------------------------|-------------------|--------------------|--------------------|----------------|--------------------------|--------------------|
| UniST [70]                   | –                 | –                  | –                  | ✓              | –                        | –                  |
| TrajFM [43]                  | ✓                 | –                  | ✓                  | ✓              | –                        | –                  |
| MobilityGPT [24]             | ✓                 | –                  | ✓                  | –              | –                        | –                  |
| UniMove [23]                 | ✓                 | –                  | ✓                  | –              | –                        | –                  |
| MoveGPT [22]                 | ✓                 | –                  | –                  | ✓              | –                        | –                  |
| Mobility-LLM [19]            | ✓                 | –                  | ✓                  | ✓              | –                        | –                  |
| MoveFM-R [47]                | ✓                 | ✓                  | ✓                  | ✓              | –                        | –                  |
| CityGPT / CityEval [14]      | ✓                 | ✓                  | –                  | ✓              | –                        | –                  |
| PMT [68]                     | –                 | –                  | ✓                  | ✓              | –                        | –                  |
| UniTraj [77]                 | ✓                 | –                  | ✓                  | ✓              | –                        | –                  |
| UrbanDiT [71]                | –                 | –                  | –                  | ✓              | –                        | –                  |
| GenMove [45]                 | ✓                 | –                  | ✓                  | ✓              | –                        | –                  |
| TrajAgent [11]               | ✓                 | ✓                  | ✓                  | –              | –                        | –                  |
| UrbanGPT [41]                | –                 | ✓                  | –                  | –              | –                        | –                  |
| UrbanVerse [60]              | –                 | –                  | –                  | ✓              | –                        | –                  |
| MoveGCL [72]                 | ✓                 | –                  | ✓                  | –              | –                        | –                  |
| ELLMob [65]                  | ✓                 | –                  | –                  | ✓              | –                        | –                  |
| Geo-Llama [37]               | ✓                 | –                  | –                  | –              | –                        | –                  |
| UrbanLLaVA / UBench [15]     | ✓                 | ✓                  | ✓                  | ✓              | –                        | –                  |
| Trajectory from Scratch [48] | ✓                 | –                  | ✓                  | ✓              | –                        | –                  |
| <b>This work</b>             | ✓                 | ✓                  | ✓                  | ✓              | ✓                        | ✓                  |

### 3.4 Controls

Five controls separate a capability failure from an artifact of the benchmark.

A *blind* condition withholds  $R$  and asks  $q_f$  alone, bounding what prior knowledge supplies without the record. In all five models, blind accuracy scores at or below the baseline for all 21 families, and is exactly zero for 8 to 18 of them depending on the model. No result below is answerable from the question alone. A *degeneracy screen* removes settings whose answer distribution leaves no headroom above the baseline. A *reference expression*, a short program we write and execute through the same pipeline the models use, scores 300 of 300, establishing that the questions are answerable. A *program* arm asks for a Python expression over the record’s points instead of a value, and executes it. A *decomposition* control hands the model the intermediate quantities instead of the record and asks for one link of the computation at a time: pick the largest of a printed list, total a printed list, or compute the distance between two supplied points. Where the other four ask whether the benchmark is sound, this one asks which step of the computation fails.

Asking for a value and asking for a program are different requests, and the difference separates two things a single accuracy number merges: whether a model can state the definition of a descriptor, which is public, and whether it can evaluate that definition on a record in front of it, which is not. We evaluate emitted programs by *extensional equivalence*: a program counts as knowledge only if it reproduces the reference on records it never saw.

### 3.5 Experimental setup

We evaluate five open-weight instruction-tuned models spanning 3.8B to 70B parameters: Phi-3.5-mini [1], Mistral-7B-Instruct-v0.3 [31], Llama-3.1-8B-Instruct and Llama-3.1-70B-Instruct [21], and Gemma-3-12B-IT [18]. All runs use greedy decoding, a single pass, and a 32,768-token context window that every record fits inside. What we measure is therefore reasoning decay rather

than context exhaustion. Every benchmark question carries the same instruction: answer with the value alone and do not explain. The instruction follows the logic of probing studies, which ask whether a model encodes a numeric quantity [63]. A short program computes each descriptor exactly, which gives every question an exact answer key, and we ask whether the model reaches that value from the record without writing the computation out. We cap generation at 24 tokens to keep a model that writes a long derivation from buying test-time compute a directly answering model never gets. Section 4.6 lifts the cap, allows code and changes the framing. Few answers reach the cap. In the full condition models generate a median of 4 to 5 tokens and hit the ceiling on at most 1.88% of questions, and almost no answer is cut off mid-string.

A generation is scored by taking the text after an Answer: marker when one is present and the last number otherwise; a generation from which no answer can be extracted is scored incorrect rather than dropped, which keeps the denominator at the full setting. Extraction fails on 0.03% to 2.18% of a model’s questions overall. On a single family it runs higher, worst for Gemma-3-12B on the retrieval probe at 18.19%. Geometric answer keys are whole numbers and we do not round onto them: a decimal scores wrong however close. Rescoring under the key’s own rounding rule raises Phi-3.5-mini from 0.0005 to 0.0037 and moves no cell across its baseline. Intervals are 95% bootstrap intervals clustered on the person whose record the question describes. The study spans 1,243 scored cells. We apply Benjamini-Hochberg [6] at  $q = 0.05$  over the 705 cells that carry an interval, Table 2 among them, and 398 survive. The test is one-sided, because every claim we correct is directional, and it reads each percentile interval as a normal approximation. No geometric cell enters the correction at all. None has a lower interval bound above its baseline before it is applied, so the result of Section 4.1 does not rest on the choice of correction. A benchmark of this shape invites reading many cells at once, a failure mode already identified for capability claims read across large task-metric-model grids [56].

## 4 Experiments

### 4.1 Main results: the recovery profile

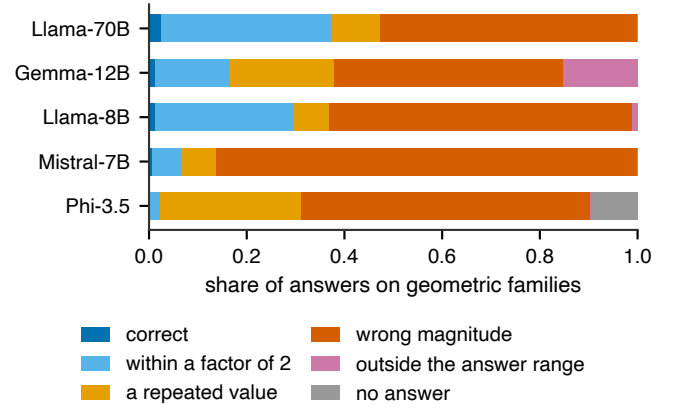
Table 2 reports the best gain each model achieves on each family over all record lengths. By class: retrieval families exceed their baselines at most lengths and are the only class that survives the longest records intact; count and sequence families clear on short records and decay; temporal families clear narrowly and early; windowed counts clear at the smallest window and, in one model, the second smallest; and spatial-geometric families clear nowhere.

Four geometric families across five models give twenty settings. Asked for the value alone, twenty of twenty never clear their baseline at any record length. Radius of gyration, maximum displacement, total distance travelled and longest single jump stay within  $+0.021$  of the modal-value baseline at every length they are generated at, 32 rows to 512. Two of the 100 settings sit above it, and no setting's interval reaches the 0.05 presence threshold. The null is therefore a measurement rather than a failure of power. None exceeds the baseline at a tolerance of  $\pm 1$  absolute or  $\pm 5$ ,  $\pm 10$  or  $\pm 25$  percent relative, each scored against its own best-constant baseline. Over the 20 geometric settings the best gain at any tolerance is  $+0.022$ , against a threshold of 0.05: loosening the metric raises the baseline at least as fast as it raises the model. Continuous error metrics agree, scoring the models worse than a constant predictor. Exact string match is nonlinear, and more than 92% of claimed emergent abilities appear under it or one other discontinuous metric and evaporate when the metric is made continuous [56]. A null that survives the continuous metric is therefore a null about the models rather than about the scoring rule.

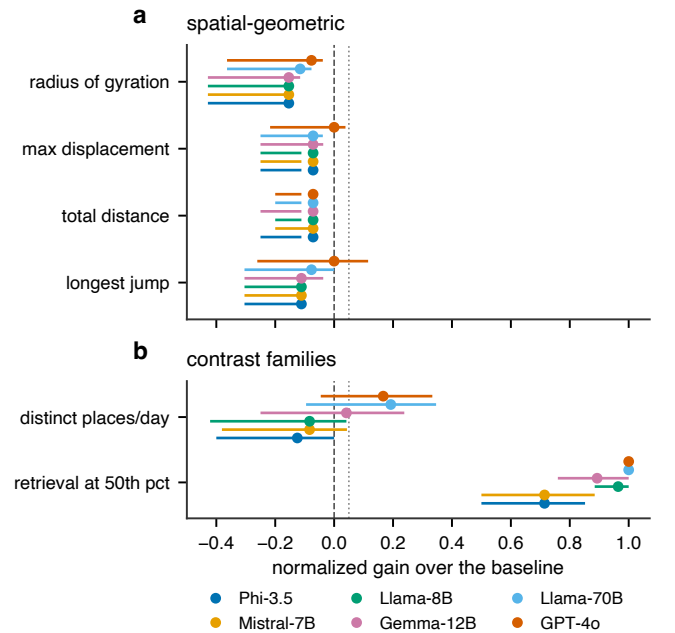
The count of recoverable families rises with scale, from 10 at 3.8B to 14 at 70B, and no geometric family moves out of the unrecoverable block at any scale. These families have no extinction length, no record length at which recovery falls back to the baseline, because exact recovery never rises above it to begin with.

The absence is specific to exact recovery. On the ordinal measure 75 of 100 geometric settings reach a rank correlation of at least .20 against the true ordering of people: a model can order individuals by these quantities while stating no individual's value. Llama-3.1-70B and Mistral-7B reach the bar on 19 of their 20 settings each, against 7 to 16 for the others. Geometry fails at naming a number. The ordering shows the quantity is not wholly absent.

*Three-level measurement.* Exact match alone would report most of this study as one undifferentiated failure. Across 190 model, family and length settings, 150 (78.9%) occupy a set of levels consistent with the three being nested, and the 40 that are not are all of one type: ordinal signal without a distributional match (Table 3). The largest single group, 93 settings, clears the distributional bar and shows a usable ordering with no exact recovery at all. Reporting accuracy alone would score those settings identically to the 16 settings that carry nothing. The distributional bar asks a model to beat a constant predictor, and a permutation of the true values beats it perfectly. Shuffling who holds which value scores zero divergence in all 190 settings while destroying every individual. Measured against that free predictor instead of the constant, the median setting sits 0.66 of the way back toward the constant, and 71 of 190 are nearer the permutation. Distributional agreement is the easiest



**Figure 2: What the models answer instead, over the four geometric families. Exact correctness is a sliver for every model, but the remainder changes with scale: answers within a factor of two rise from 2.2% at 3.8B to 34.8% at 70B, while the smallest repeats one value on 28.8% of questions.**



**Figure 3: A frontier model on the same items: median normalized gain with 95% intervals, six models, 30 questions per cell. The dotted line marks the presence threshold. GPT-4o leads on the contrast families and clears no geometric cell.**

of the three levels to obtain and the least informative about any person.

The amount of composition an answer requires does not predict which families survive. Single-lookup and multi-step families appear on both sides of the boundary. The profile is organised by what kind of quantity is asked for rather than by how much work the answer takes.

**Table 2: Best normalized gain over the baseline that each model reaches on each descriptor family, over every length at which that family exists: seven for distinct places per day and recency, five for the geometric families, fewer for the longer windows. Bold marks a family whose 95% interval excludes the baseline at some length. No geometric family exceeds the baseline for any model at any length, while every count and sequence family clears it for at least one.**

| Descriptor family                    | Phi-3.5       | Mistral-7B    | Llama-8B      | Gemma-12B     | Llama-70B     |
|--------------------------------------|---------------|---------------|---------------|---------------|---------------|
| <i>Retrieval by position</i>         |               |               |               |               |               |
| retrieve at 10th pct                 | <b>+0.919</b> | <b>+0.959</b> | <b>+0.986</b> | <b>+0.993</b> | <b>+1.000</b> |
| retrieve at 30th pct                 | <b>+0.889</b> | <b>+0.872</b> | <b>+0.973</b> | <b>+0.993</b> | <b>+0.990</b> |
| retrieve at 50th pct                 | <b>+0.905</b> | <b>+0.807</b> | <b>+0.986</b> | <b>+0.976</b> | <b>+0.986</b> |
| retrieve at 70th pct                 | <b>+0.949</b> | <b>+0.881</b> | <b>+0.990</b> | <b>+0.993</b> | <b>+0.993</b> |
| retrieve at 90th pct                 | <b>+0.956</b> | <b>+0.847</b> | <b>+0.990</b> | <b>+0.997</b> | <b>+0.997</b> |
| retrieve probe                       | +0.047        | <b>+0.123</b> | <b>+0.847</b> | <b>+0.953</b> | <b>+0.977</b> |
| <i>Count and sequence</i>            |               |               |               |               |               |
| distinct places per day              | <b>+0.086</b> | <b>+0.073</b> | <b>+0.112</b> | <b>+0.323</b> | <b>+0.737</b> |
| two-hop path                         | <b>+0.730</b> | <b>+0.267</b> | <b>+0.801</b> | <b>+0.588</b> | <b>+0.932</b> |
| transition                           | <b>+0.447</b> | <b>+0.719</b> | <b>+0.776</b> | <b>+0.614</b> | <b>+0.885</b> |
| core set                             | -0.144        | -0.091        | -0.023        | +0.061        | <b>+0.103</b> |
| <i>Temporal</i>                      |               |               |               |               |               |
| longest stay                         | -0.098        | -0.028        | -0.127        | -0.087        | +0.005        |
| recency                              | <b>+0.220</b> | <b>+0.258</b> | <b>+0.386</b> | <b>+0.251</b> | <b>+0.810</b> |
| modal timeslot                       | <b>+0.331</b> | <b>+0.351</b> | <b>+0.470</b> | <b>+0.422</b> | <b>+0.557</b> |
| <i>Windowed count (state volume)</i> |               |               |               |               |               |
| distinct in last 16                  | -0.015        | +0.026        | <b>+0.093</b> | <b>+0.123</b> | <b>+0.134</b> |
| distinct in last 32                  | -0.018        | +0.025        | -0.014        | +0.014        | <b>+0.050</b> |
| distinct in last 64                  | -0.021        | -0.024        | +0.003        | -0.031        | +0.000        |
| distinct in last 128                 | -0.024        | -0.007        | +0.000        | -0.031        | +0.003        |
| <i>Spatial-geometric</i>             |               |               |               |               |               |
| radius of gyration                   | -0.099        | -0.086        | -0.092        | -0.095        | -0.041        |
| max displacement                     | -0.020        | -0.020        | +0.014        | +0.000        | +0.021        |
| total distance                       | -0.007        | -0.007        | -0.003        | -0.007        | -0.003        |
| longest jump                         | -0.042        | -0.042        | -0.010        | -0.024        | -0.017        |

The wrong answers also have a shape that changes with scale (Figure 2), and exact correctness stays between 0.1% and 2.6% for all five models. Two failure modes sit behind a single null: a constant that carries no information about the person, and an answer that varies with the record but is the wrong size.

*A frontier model.* The five models above are open-weight and none exceeds 70B, so the geometric null could be read as a property of that range. We score GPT-4o [50] on 960 questions, 30 per cell over the four geometric families and two contrast families, distinct places per day and retrieval at the median position, through the same parser and per-setting baselines, and re-score the five open-weight models on those same items so every column carries the same denominator. GPT-4o is the strongest model in the comparison, with a median gain of +0.000 across 32 cells against -0.054 for Llama-3.1-70B and the retrieval family at 1.000 at all five lengths, and it clears no geometric cell (Figure 3). Its best estimate, maximum displacement at 32 rows, is +0.107 with an interval of [-0.131, +0.192] that spans the threshold, and it states total distance correctly on none of the 150 questions it is asked.

## 4.2 Which step of the computation fails

The geometric result says a descriptor does not come back. It does not say which part of producing it breaks. We hand the models the intermediate quantities directly and ask for one link at a time,

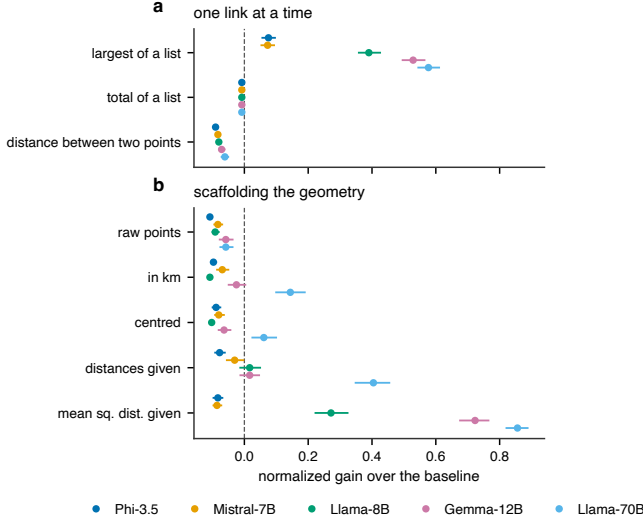
scoring each at exact match and, because these are continuous quantities, also within a factor of two of the truth. Figure 4(a) reports all three with their intervals.

Handed a printed list and asked for its largest element, every model clears its baseline and is within a factor of two of the answer on at least 99% of items. The pipeline that produces the geometric nulls therefore produces a success on request, through the identical prompt, parser, decoder and scorer.

Handed a printed list of distances and asked for the total, no model states the exact answer at any size, but within a factor of two they score between .619 and 1.000, and the two Llama models are within a factor of two on every item. Aggregation is intact and exact statement of the total is not, an arithmetic-precision limit rather than an inability to combine.

Handed two points and asked for the distance between them, the models fail at both tolerances. Exact recovery reaches at most 2.8%, and within a factor of two the best model reaches 39.5% while the rest sit between 8.8% and 13.7%. This is the only one of the three that fails when the requirement to be exact is removed.

The three controls together locate the failure. It is not the harness, which succeeds on demand. It is not aggregation, which survives at tolerance. It is the geometric step itself, which fails even when the points are supplied and even when approximate answers are



**Figure 4: Normalized gain with 95% intervals for the decomposition controls (a) and the scaffold ladder (b), a five-rung sequence handing the model ever more of the geometric computation. A point right of the dashed line beat the best-constant predictor. Every model clears on the largest of a list; none clears on the distance between two points. The ladder recovers geometry only once squared distances are supplied.**

accepted. That also explains why the program condition of Section 4.7 recovers so much. A model that cannot carry out the step can still name it correctly in code.

### 4.3 Effect of record length

Record length does not act on the profile the way a decay curve would suggest. Across the 105 model-family pairs the study covers, 47 (45%) exceed their baseline by the interval bar at the longest record that family is generated at, 46 (44%) never clear it at any length, and only 12 (11%) clear it and then stop. Graded decay is the minority behaviour. For most pairs, the descriptor either works everywhere we look or nowhere. The split is not a restatement of the geometric null. Excluding the geometric families it is 86% of 85 pairs, and excluding the longest windows as well it is 84% of 75.

The 12 pairs that do have an extinction length spread it across the whole range, from 8 rows to 256, and the same family extinguishes at different lengths in different models: the count of distinct places per day survives to 128 rows in two models, to 64 in a third and only to 8 in Mistral-7B. An extinction length is a property of a model-family pair rather than of the descriptor.

Within those pairs the decline is steep. At 70B over the common window from 16 to 512 rows, two-hop paths fall from +0.932 to +0.152, transitions from +0.885 to +0.431, and distinct places per day from +0.466 to +0.079. Figure 5 puts these against the geometric families, whose lines stay flat at the baseline instead of descending. A family that never had a signal produces the same endpoint as one that lost it, and only the curve’s shape distinguishes them.

Heterogeneous length behaviour has been observed before. Du et al. [12] report degradation of 13.9% to 85% across tasks even

when retrieval is perfect, in a single table of closed-model results. Measuring every family at every length it exists at, against an exact baseline, turns those instances into a partition of the space.

Mean accuracy over this grid lands where no pair lives: 45% of pairs clear at the longest record, 44% never clear, and the average of the two describes neither. A benchmark that scores one record length also sees none of the 11% that move.

*Required state.* A windowed family asks how many distinct places appear in the last  $W$  rows, and its answer depends on  $W$  items however long the record is. We cross that required state against the length of the surrounding record. At fixed required state, rank correlation falls as the record grows, by  $-0.414$ ,  $-0.464$  and  $-0.231$  for required states of 16, 32 and 64 items. At fixed record length it does not fall as required state grows. The task requires exactly 16 items of state in every one of those settings, and the capability still decays by .414 as the record around those 16 items gets longer.

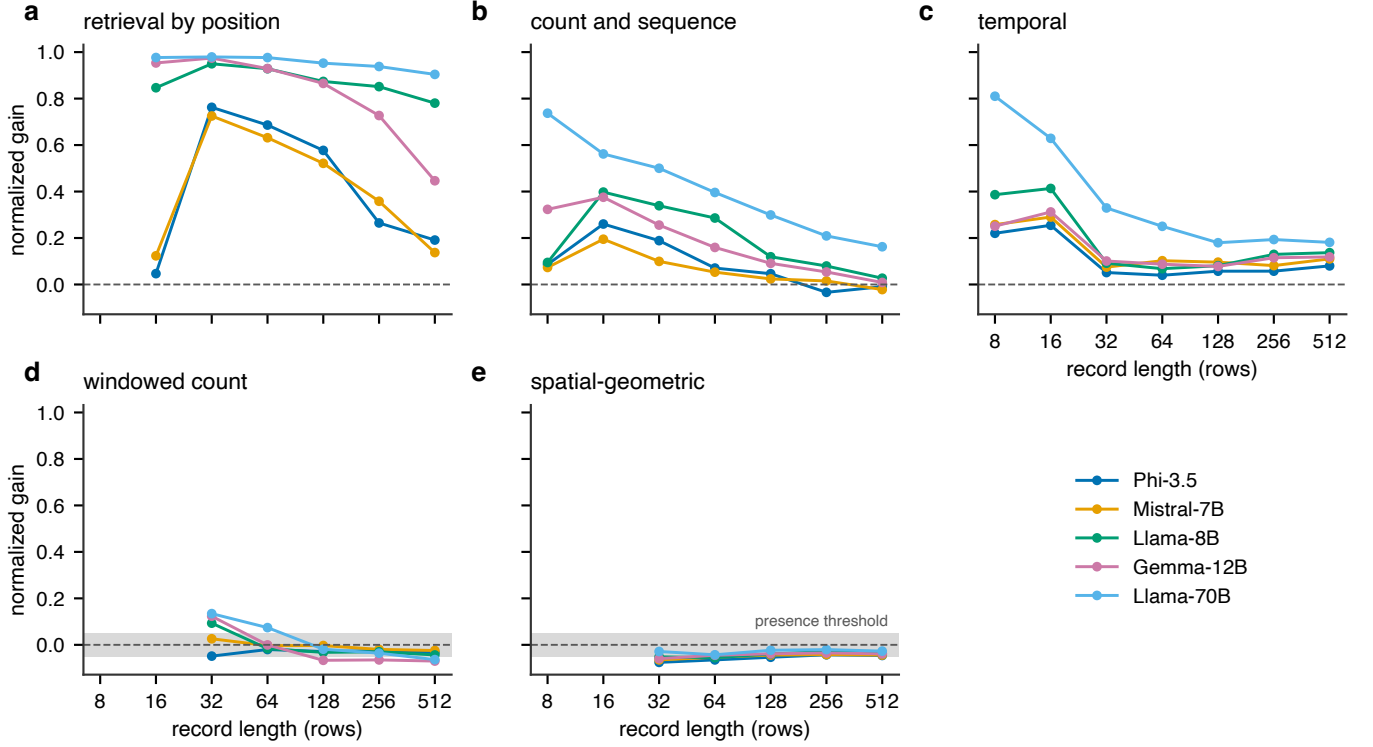
A tail-location control rules out the alternative that the model cannot find the end of the record. A pure locate task near the record’s end scores .997, .980 and .953 at 32, 64 and 128 rows, across the range where rank correlation falls from .808 to .494. The record’s size binds recovery, and the answer’s difficulty does not.

*Answer position.* Record length binds recovery. We next ask whether the answer’s location in the record explains what remains. Answer position affects recovery in some models and not others, in both directions, and the effect is not ordered by scale. At 512 rows Llama-3.1-70B scores .927 with the answer at the record’s tenth percentile and .970 at the seventieth, with intervals [.897, .953] and [.95, .987] that overlap, while another model spreads .423 across positions at the same length. No onset survives a ceiling control. Veseli et al. [62] report the lost-in-the-middle effect disappearing above half the context window with a recency bias remaining, and Liu et al. [44] established the effect at ordinary context lengths. Position is a per-model property on this benchmark and does not explain the recovery profile.

*Absence detection.* Every result so far asks the model to produce a value the record contains. A record can also fail to contain the answer, and reporting that is a separate capability. We pair unanswerable questions with answerable ones at the same record length and score both, in all five models.

Abstention splits the models in two (Figure 6, appendix). Three of the five start able to report absence and lose it as records grow. Correct abstention falls from .97 to .13 in Gemma-3-12B, .99 to .66 in Llama-3.1-70B and .85 to .52 in Llama-3.1-8B, with non-overlapping intervals at the endpoints in each. The other two never had it. Mistral-7B peaks at .16 at the shortest record and reaches .00 by 512 rows, and Phi-3.5-mini never exceeds .10 at any length. For these two there is no collapse to measure. Only the three models at 8B parameters and above show the capability at all.

Where abstention does collapse, accuracy on the *answerable* half holds. Llama-3.1-70B falls from .98 to .94 across the same range while its abstention loses a third, and Gemma-3-12B answers 96% of answerable questions correctly at 128 rows while abstaining correctly on 54% of unanswerable ones. The false-abstention rate, the share of *answerable* questions refused, stays at or below 3% for



**Figure 5: Normalized gain against record length for all five models, averaged over the families in each descriptor class: retrieval (a), count and sequence (b), temporal (c), windowed counts (d), spatial-geometric (e). Shading marks the presence threshold. The classes separate the same way in every model, and no model lifts the geometric families out of the threshold at any length.**

four of the five models and peaks at 10.7% for Llama-3.1-8B, well below the collapse in correct abstention over the same range.

That control identifies the failure. A model growing generally more cautious would refuse answerable questions too, and these do not. The model instead fabricates a specific answer to a question whose answer is absent, more often the longer the record. The same qualitative pattern is reported for two of the models in a multimodal needle search as haystacks grow [64]. Our paired design reports both halves at the same record length. The answerable half is still at .593 where abstention has reached .130.

#### 4.4 Cross-record attribution

Cross-record attribution shows the model  $K$  people’s records and asks which of them visited a given place on a given day. It succeeds, and it improves as the same rows are split across more records. Holding total length fixed at roughly 440 rows and splitting it  $K$  ways, Llama-3.1-70B rises from +0.705 at  $K = 2$  to +0.909 at  $K = 8$ , with raw accuracy rising from .803 to .919. The gain is therefore not an artifact of the falling baseline. More records means more distractors, which should make an attribution join harder. It does the opposite, because splitting the same rows more ways leaves fewer rows per record. This reaches Section 4.3’s conclusion by shortening each record without removing rows.

Two models qualify this. Mistral-7B and Phi-3.5-mini rise from a baseline-level +0.040 and +0.055 at  $K = 2$  to +0.401 and +0.313

**Table 3: Which of the three measurement levels each model-family-length cell occupies, over the seven families whose answers are magnitudes. The remaining fourteen return place identifiers, for which a distributional match and a rank correlation are not defined, and they carry no level below exact. 150 of 190 cells (78.9%) are consistent with the levels being nested; every violation is of one type, ordinal signal without a distributional match. Exact match alone would collapse the 166 cells without exact recovery into one undifferentiated failure.**

| Pattern | Levels present                        | Cells | Nesting          |
|---------|---------------------------------------|-------|------------------|
| D0-     | distributional + ordinal, no exact    | 93    | nested           |
| D-      | distributional only                   | 17    | nested           |
| -       | no level present                      | 16    | nested           |
| D0E     | all three levels                      | 24    | nested           |
| -0-     | <b>ordinal without distributional</b> | 40    | violates nesting |

at  $K = 6$ , clearing their baselines at  $K = 4, 6$  and 8, then dip slightly at  $K = 8$ . The monotone claim therefore covers three of five models and the direction covers all five.

#### 4.5 Effect of elicitation

Sections 4.3 and 4.4 locate the constraint in the record, which leaves open whether the profile is a property of the descriptors or of how

we asked. Three interventions move the level of recovery and leave the split by descriptor class intact.

Wording moves little, and least where there is least to move. Across the twenty geometric settings, five phrasings of the same question produce a spread with median .008 and maximum .044, but every accuracy there lies within .068 of zero, which makes the small spread partly a floor. On the ten settings that carry signal the spread is larger, median .030 and maximum .137, which bounds the wording effect without dismissing it.

Scaffolding helps where capacity exists, and how much it helps is graded by capacity rather than switched on by it. A five-rung ladder hands the model progressively more of the computation, ending at a rung where only a square root remains. On that final rung Llama-3.1-70B achieves +0.856, Gemma-3-12B +0.723 and Llama-3.1-8B +0.271, all with intervals excluding the baseline, while Mistral-7B and Phi-3.5-mini stay below it at −0.086 and −0.083. The first rung alone buys +0.202 at 70B, with an interval excluding the rung below.

Changing the answer format recovers graded signal and then loses it with length. Asking which of two records has the larger value, instead of asking for the value, discriminates at 70B on short records down to a ratio of 1.46, and three of four geometric families lose the .75 discrimination threshold by 512 rows. Sampling and aggregating eight draws rescues one setting of eight.

#### 4.6 Mobility framing, reasoning and tools

Three conditions each change one element of the protocol: how the record is framed, how much room the model has to reason, and whether it can use code (Appendix E).

Geometric recovery fails whether the same coordinates are framed as a person’s location record, as a set of numbered points, or as bare  $x, y$  coordinates. No setting clears its baseline under any framing, and the largest shift between framings is 1.1 points.

More room to reason recovers one descriptor in one model. Doubling a chain-of-thought [66] budget from 4,096 to 8,192 tokens leaves Llama-3.1-8B at .018 and .020 exact accuracy, at or below its baseline in every setting. With 8,192 tokens, Llama-3.1-70B recovers the longest jump on records of 32 to 128 rows, the three settings that survive a Benjamini-Hochberg correction. Its total distance never exceeds the baseline, even when the reasoning finishes, consistent with the precision limit on totals in Section 4.2.

A tool recovers the values in Llama-3.1-70B alone, .963 exact against .046 when it states them. The failure therefore lies in the geometric computation: the framing changes no verdict, and only the largest model recovers values by writing the computation out.

#### 4.7 Specifying the computation instead of performing it

Asked for a Python expression over the record’s points instead of a value, the models produce programs we can execute and compare against the reference. We score them by extensional equivalence: a program counts only if it reproduces the reference computation on 50 records it never saw.

The arm separates knowledge from luck, and Table 4 gives it per family. Each model is asked for 1,200 expressions, 300 in each of the four geometric families. Llama-3.1-70B produces 1,155 that are correct on the record they were written for, and all 1,155 remain

**Table 4: Program arm. Each cell is expressions that reproduce the reference on 50 held-out records, over expressions correct on the model’s own record. Gemma-12B holds on two of the four families, and the hold rate is not ordered by capacity: Phi-3.5 keeps 49 of 50 while Llama-8B keeps none of 79.**

| model      | $r_g$   | max disp. | total dist. | long. jump | all       |
|------------|---------|-----------|-------------|------------|-----------|
| Phi-3.5    | 0/0     | 34/35     | 0/0         | 15/15      | 49/50     |
| Mistral-7B | 0/0     | 0/0       | 0/0         | 0/0        | 0/0       |
| Llama-8B   | 0/56    | 0/0       | 0/0         | 0/23       | 0/79      |
| Gemma-12B  | 0/0     | 34/34     | 0/0         | 84/84      | 118/118   |
| Llama-70B  | 255/255 | 300/300   | 300/300     | 300/300    | 1155/1155 |

correct on the held-out records, 96.2% of what it was asked for. Gemma-3-12B produces 118 and holds all 118, and Phi-3.5-mini produces 50 and holds 49: 9.8% and 4.1% of what each was asked for. Their programs survive when they appear but appear rarely. Llama-3.1-8B produces 79 expressions that are correct on their own record and *zero* survive the held-out test. Correctness on the visible record alone certifies nothing. Mistral-7B emits parseable code on 6% of items and produces none that run, which bounds what this arm can say about that model rather than about the descriptor.

How much a model produces rises with capacity. Whether what it produces survives does not. Phi-3.5-mini keeps 49 of 50 at 3.8B while Llama-3.1-8B keeps none of 79. Volume and survival are different quantities and only the first is ordered by scale.

Llama-3.1-70B clears this bar on all four geometric families. Gemma-3-12B clears it on two, maximum displacement and longest jump, and writes nothing that survives for radius of gyration or total distance. Phi-3.5-mini clears the same two at 3.8B. Where a model does clear it, the failure to state the value is not ignorance of the descriptor. It is a failure to carry the computation out, and our evidence for that covers three of the five models on the families named here rather than all five on all four.

## 5 Discussion

*The rank-level dissociation.* The field scores generated mobility by comparing descriptor distributions against real ones, and Section 4.1 finds the recovery direction empty for exactly the descriptors those scores are built from. Permuting the true values across people leaves every distributional score untouched while destroying every individual: a predictor holding no information about anyone reproduces the population exactly. Asked what share of people exceed the population’s own 75th percentile, where truth and permutation both give 25.0%, the five models answer between 0.0% and 87.3%. Yet the same answers still order people. On the four geometric families Llama-3.1-70B reaches a median normalized gain of +0.204 at selecting the top decile, with intervals excluding zero in 10 of 20 settings, while its exact recovery clears its baseline in none. Table 5 in the appendix gives both columns for all five models. The same answers that never clear the exact bar carry a positive rank gain in four of them. Halving every true value preserves every rank by construction while the count falls to a median of 2.0%, placing the failure in the level rather than in the ordering. A system that ranks people by these descriptors can use a model that cannot state one; a system that counts people against a stated threshold cannot.

*Mitigation arms.* Eight interventions act on geometric recovery and four move it. Scaffolding short of performing the geometric step for the model, relaxing the metric, rephrasing the question and aggregating repeated samples all leave the geometric families where they were. Asking for a program succeeds in three of the five models, asking for a ranking on short records, and allowing a tool call in one. None of the three requires stating the value. Chain-of-thought [66] is the one intervention that recovers a stated value: the longest jump, in Llama-3.1-70B, on short records (Section 4.6).

*Elicitation.* The decomposition control, the program arm and the scaffold ladder each place the failure in the geometric step (Sections 4.2, 4.7 and 4.5), and Section 4.6 shows the same failure when the record and question carry no mobility terms.

*The program arm.* Section 4.7 shows three of the five models specify computations they cannot perform, on the same geometric families whose values they cannot state. Huang et al. recommend exactly this route for direction estimation, an agent calling an external API or writing a Python script, and do not test it [28]. We test it. Systems that need a descriptor from a record have a route that does not depend on the model computing it, at the capacities where the programs survive an equivalence test.

*Future work.* The ordinal signal that geometric families carry suggests asking for rankings where a ranking is enough. Whether the far longer traces of reasoning-trained models achieve these descriptors is a question for a different generation regime. A human baseline would separate a task these models cannot do from one that is harder than it looks. The reference expression and the program arm bound answerability without measuring it.

## 6 Limitations and scope

*Measurement.* Every result outside the program arm and Section 4.6 comes from one elicitation regime. The model answers with the value alone under a 24-token cap. No model derives an answer before stating it, and those nulls are nulls about direct statement. Llama-3.1-70B reasons at one budget, and the tool condition stops at 128 rows. Because every result except the two-model sampling arm uses greedy decoding and a single pass, we report no seed or sampling variance for the main benchmark. The scaffold ladder and the wording study ran the full condition only, and neither carries its own record-free control. The comparison arm’s positive control clears in only 4 of 25 cells, and in none for the three largest models, whose answers on that family go flat only at the longest records. Their geometric discrimination result therefore rests on the internal validity of their own settings rather than on the control.

*Contamination.* The substrate is public, but every question is constructed. To use a memorised table a model must first identify the record from coordinates it has never seen in that orientation, and a rotation-invariant signature good enough to do that is the aggregate computation the task asks for.

*Scope.* We evaluate one substrate and run no cross-dataset replication. We evaluate five open-weight models from 3.8B to 70B, and one API model on 30 questions per cell over six of the 21 families. The mobility foundation models of Table 1 are not evaluation targets. Fifteen of the twenty expose no interface that accepts a record-level

question. UrbanGPT [41] accepts questions but takes aggregated urban sensing data as input and has no individual record to ask about. We report no human baseline. The reference expression and program control carry the answerability argument.

*Ethics and data.* The substrate is YJMob100K [69], a publicly released dataset of anonymised human mobility collected from mobile devices. Locations are discretised to a grid and the city is withheld by the data provider, and records carry no street-level position and no place names. We report distances in grid units converted to kilometres, and we name, map and geocode no location. A per-question rigid transform and relabelling from a pool of 512 identifiers change the coordinates a model sees without changing the descriptor’s value. The questions we generate describe individual movement patterns, and we release no per-person output. Code that rebuilds every question from YJMob100K and scores model answers is available at <https://github.com/mustafasameen/GAUGE>. The risk this work carries is that a method for recovering summary quantities from one person’s record is also a method for profiling that person. Our results bear on the shape of that risk rather than removing it. When stating the value directly, no model recovers the spatial-geometric descriptors exactly at any length we test. Four results cut the other way. Count and sequence descriptors clear on short records, the models order individuals on geometric quantities in 75 of 100 settings, and three of the five write programs that compute those quantities on records they never saw. With chain-of-thought reasoning, the largest model also recovers the longest jump. The geometric result is a limit on stating a value directly from a record, and not evidence that these quantities are out of reach. We report aggregate accuracies only, and the transforms we apply mean an answer cannot be traced to a person in the source data.

## 7 Conclusion

We asked whether language models can recover mobility science’s descriptors from a single person’s location record, using a benchmark of 34,200 generated questions. Recovery splits by descriptor class. Asked to state the value, five of five models never clear the baseline on a geometric descriptor, at any record length from 32 to 512 rows. The other classes do not simply decay either. Across the 105 model-family pairs, 47 clear their baseline at the longest record we test and 46 clear it nowhere: length usually decides whether a descriptor is recoverable at all. The binding quantity is the amount of record rather than the state the answer requires. Two designs that turn different variables agree. The failure is not ignorance: three of the five write programs that compute these quantities correctly on records they never saw. The failure also holds without mobility terms in record or question, and chain-of-thought reasoning recovers only the longest jump, in the largest model, on short records. Recovery from one person’s record gives mobility foundation models a second evaluation axis beside the distributional scores in use, and GAUGE measures it.

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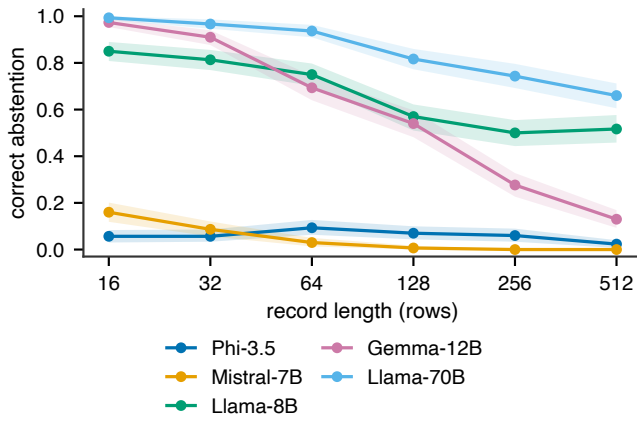
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**Table 5: Two decisions from one set of answers, on the four geometric families. Exact gain is normalized gain over the per-setting baseline. Rank gain positions top-decile selection between a permutation of the true values, which carries no information about any individual, and the true values themselves. The count is the share of people placed above the population’s true 75th percentile, where both the truth and a permutation give 25%.**

| model      | exact gain | rank gain | CI>0  | count (%) |
|------------|------------|-----------|-------|-----------|
| Phi-3.5    | -0.040     | +0.078    | 2/15  | 0–2       |
| Mistral-7B | -0.040     | +0.071    | 0/20  | 0–20      |
| Llama-8B   | -0.019     | +0.058    | 4/20  | 0–71      |
| Gemma-12B  | -0.023     | -0.001    | 1/20  | 0–87      |
| Llama-70B  | -0.019     | +0.204    | 10/20 | 0–40      |



**Figure 6: Absence detection against record length for the five models: correct abstention on unanswerable questions. Three models start able to report absence and lose it as records grow, Gemma-3-12B falling from .973 to .130, and the two smallest never abstain. Bands are 95% intervals.**

**Table 6: Summary of the 34,200-question set behind the recovery profile.**

| statistic                 | value                             |
|---------------------------|-----------------------------------|
| questions                 | 34,200                            |
| descriptor families       | 21                                |
| record lengths            | 8, 16, 32, 64, 128, 256, 512 rows |
| distinct people           | 16,388                            |
| family-by-length settings | 108                               |
| questions per setting     | 300 (600 for the retrieval probe) |
| source corpus             | YJMob100K                         |

## A Benchmark statistics

Table 6 summarises the 34,200-question set behind the recovery profile. With questions drawn from 16,388 distinct people, no record carries a meaningful share of any cell. The 34,200 questions split 28,200 over place identifiers and 6,000 over coordinates, the four spatial-geometric families.

## B Prompt form

Every question presents the record under a header naming its columns, followed by one row per observation, followed by the question. Place-family records render a place identifier per row and coordinate-family records render a coordinate pair. One instruction precedes every question, in the user turn and identical across models and families:

You answer questions about location records. Reply with ONLY the value, on a single line of the form ‘Answer: <value>’. Do not explain. If you cannot determine the value, reply ‘Answer: unknown’.

The two record forms are:

Here is a person’s location record. Each line is day, timeslot, place.  
d40 t18 p144  
d40 t19 p144

...  
Question: What is the last place this person was at STRICTLY BEFORE day 40 timeslot 24? Answer with the place number.  
Answer with the value only.

Here is a person’s location record. Each line is day, timeslot, place.  
d37 t18 x-341 y-387  
d37 t19 x-342 y-386

...  
Question: Treating each unit on the x and y axes as 500 metres, what is the radius of gyration of this person’s locations, in kilometres? The radius of gyration is the root-mean-square distance of the visited points from their mean position. Answer with a whole number of kilometres.  
Answer with the value only.

Generation stops at 24 tokens. In the blind condition Mistral-7B reaches that ceiling on 23.50% of questions, and every other model and condition stays under 2%. We run all five models through Hugging Face Transformers [67].

The coordinate families are rendered under two column headers, day, timeslot, place and day, timeslot, x, y, with everything else held fixed. All five models are scored under both, and the reported numbers use the first. Across 100 model, family and length cells the normalized gain moves by between  $-0.031$  and  $+0.012$  and no cell changes its verdict. The answer keys are identical, which makes the comparison paired, and no geometric cell clears its baseline under either header. Because the blind condition removes the record and leaves the header and question unchanged, the baseline a model is scored against inherits the full condition’s wording.

## C Reference computation and answer checking

Every answer is checked against a value computed from the same rows the model is shown, by a program rather than by annotation. For the radius of gyration that program is

```
P = np.array(points)
r_g = 0.5 * np.sqrt(((P - P.mean(0))**2).sum(1).mean())
```

where points are the coordinate pairs parsed from the rendered record and 0.5 converts the 500-metre grid unit to kilometres. On the 32-row record printed above this returns 2.587, which the whole-number key of Section 3.5 records as 3. The key is therefore exact rather than estimated, and it is computed from the transformed, rounded points the model sees rather than from the underlying trajectory. The per-question rigid transform cannot put the answer out of reach of the record. The same program is the reference for Section 4.7. A model-written expression counts only when it returns this value on 50 held-out records.

**Table 7: Questions generated per descriptor family and record length. Empty cells were never generated or scored (Appendix D).**

| class                 | family                  | 8   | 16    | 32    | 64    | 128   | 256   | 512   | total  |
|-----------------------|-------------------------|-----|-------|-------|-------|-------|-------|-------|--------|
| Retrieval by position | retrieve at 10th pct    | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | retrieve at 30th pct    | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | retrieve at 50th pct    | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | retrieve at 70th pct    | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | retrieve at 90th pct    | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | retrieve probe          | –   | 600   | 600   | 600   | 600   | 600   | 600   | 3,600  |
| Count and sequence    | distinct places per day | 300 | 300   | 300   | 300   | 300   | 300   | 300   | 2,100  |
|                       | two-hop path            | –   | 300   | 300   | 300   | 300   | 300   | 300   | 1,800  |
|                       | transition              | –   | 300   | 300   | 300   | 300   | 300   | 300   | 1,800  |
|                       | core set                | –   | 300   | 300   | 300   | 300   | 300   | 300   | 1,800  |
| Temporal              | longest stay            | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | recency                 | 300 | 300   | 300   | 300   | 300   | 300   | 300   | 2,100  |
|                       | modal timeslot          | –   | 300   | 300   | 300   | 300   | 300   | 300   | 1,800  |
| Windowed count        | distinct in last 16     | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | distinct in last 32     | –   | –     | –     | 300   | 300   | 300   | 300   | 1,200  |
|                       | distinct in last 64     | –   | –     | –     | –     | 300   | 300   | 300   | 900    |
|                       | distinct in last 128    | –   | –     | –     | –     | –     | 300   | 300   | 600    |
| Spatial-geometric     | radius of gyration      | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | max displacement        | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | total distance          | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
|                       | longest jump            | –   | –     | 300   | 300   | 300   | 300   | 300   | 1,500  |
| total                 |                         | 600 | 2,400 | 5,700 | 6,000 | 6,300 | 6,600 | 6,600 | 34,200 |

## D Coverage grid

Table 7 gives the count of questions at every combination of descriptor family and record length. A family appears only at record lengths where its quantity is defined. Geometric families begin at 32 rows, because radius of gyration over eight rows is not a quantity a reader would recognise as a person’s range, and windowed counts exist only where the window fits inside the record. Empty cells were never generated and are never scored. A mean over all families at a fixed length is therefore computed over a different family set at each length. Every length comparison in Section 4.3 holds the family fixed, and every family comparison holds the length fixed. The 108 populated cells, not the  $21 \times 7$  product, are the unit of the study.

## E Framing, reasoning and tool conditions

*Framing.* Each of 1,500 geometric questions per model appears under three framings with identical coordinates, row order, answer key and instruction (Appendix B). The mobility framing opens “Here is a person’s location record. Each line is day, timeslot, x, y.” Relabelling changes the opening to “Here is a set of numbered points” and refers to points throughout, and stripping opens “Here is a set of points. Each line is x, y.” and drops the day and timeslot from every row. The mobility and relabelled framings run together, as do the relabelled and stripped framings, and the relabelled framing returns identical answers in both runs for every model. The largest

shifts are 1.1 points in Llama-3.1-70B, higher under the mobility framing than under the relabelled framing (95% interval 0.5 to 1.8), and 0.7 points in Gemma-3-12B, higher under the stripped framing than under the relabelled framing (95% interval 0.1 to 1.4).

*Reasoning.* Reasoning uses 400 of the geometric questions, 20 per setting, with a prompt that asks for step-by-step work ending on a line Answer : followed by the value. When a trace reaches the token limit without that line, Answer : is appended and the model generates up to eight more tokens, which gives every question an answer. Answers produced this way score below the rest in every family. Llama-3.1-8B answers the same questions with budgets of 4,096 and 8,192 tokens, and Llama-3.1-70B with 8,192. The Benjamini-Hochberg correction runs at  $q = .05$  over the 20 settings, with a one-sided binomial test of accuracy against each setting’s baseline. On total distance, the 40 traces of Llama-3.1-70B that run to completion land within a factor of two of the truth 38 times and on it once.

*Tool.* The tool condition presents each record as a Python list of points, at 32 to 128 rows, with 97 to 100 questions per setting drawn from 289 people, 72 of whom also appear in the benchmark’s geometric questions. The model writes one expression over the list and we execute it. The paired comparison is the same model stating the value for the same question. The four smaller models score .100 or less with the tool, and when their code runs, it returns the right value at most 18% of the time.